Verify the Institution, Not the Agents: A Bounded-Verified Separation-of-Powers Kernel Stress-Tested Against Colluding LLM Swarms

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Track: 4, Multi-Agent Oversight and Control (cross-track: 2, 3)

Abstract

Per-agent alignment and monitoring do not scale to swarms, and a colluding subgroup is designed to defeat both. We ask whether safety guarantees can instead live in an *institution*: code whose rules are checked once and enforced on every agent.

We build a separation-of-powers kernel. A worker proposes a release and a kernel-seated committee attests with per-auditor keys. The executive commits only if mechanical checks pass (integrity, seating, binding, quorum, policy, uniqueness), logging each release atomically. Bounded exhaustive exploration of the real implementation (23,376 states) confirms its safety, liveness, at-most-once and audit invariants, and catches all seven deliberately broken variants. The resulting guarantee is conditional: an unsafe release needs k false attestations from seated auditors.

We stress-tested eight governance regimes with 32,830 completed calls to two reasoning-model families, in 50-agent swarms where hidden cells (0â€“50%) pushed invalid releases.

- **Mechanical checks were decisive.** Plain majority voting approved every forged-signature release (48/48 per model). Every kernel regime blocked them all, including a mechanically gated majority with no committee.
- **Committee capture came only from colluder seats** and never exceeded the bound set by the seats actually drawn. A gated majority was safer; committees need far fewer model requests (7 vs 50) but are capturable.
- **Honest agents almost never erred,** even on near-miss receipts, so the independence assumption behind committee bounds remains untested.
- **Colluders' outputs ran 1.9â€“2.8Ã— longer,** an association worth testing as a monitoring signal.

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1. Introduction

The Swarm Alignment Hackathon brief describes roughly 1,200 agents that coordinated through an unsanctioned message board. Its point is that no single-model safety test would have caught this. Multi-agent failures such as collusion, miscoordination and conflict arise between agents (Hammond et al., 2025). Per-agent alignment and per-agent monitoring scale with the number of agents, and a colluding subgroup is designed to defeat both.

Human institutions take a different route. We do not verify every citizen. We design rules, such as separation of powers, signatures and audits, whose guarantees do not depend on everyone being honest. Recent work proposes separation of powers for agent economies (*From Logic Monopoly to Social Contract*,

arXiv:2603.25100). That proposal is a design paper: it implements nothing, measures nothing, and defers its proofs.

An agent institution, unlike a human one, is code. Its rules can be **mechanically verified** and **stress-tested** against real AI swarms with planted adversaries, two things no constitution allows.

Theory of change. Deployers already route agent actions through gateways (MCP, A2A, LLM proxies). If consequential actions must pass an institution kernel at that gateway, the kernel's mechanical guarantees hold regardless of swarm size or model. What remains is to measure the few behavioural parameters the guarantee depends on: honest error, its correlation, and colluder compliance.

We show this loop of build, verify, stress-test and measure is feasible in a weekend. A fellowship would extend it to tasks where honest agents genuinely err, to verifiable seating, and to deployment at a real gateway.

Contributions.

1. A small separation-of-powers kernel for agent swarms. It splits what can be checked mechanically (integrity, seating, binding, quorum, policy, uniqueness, audit) from what must be judged (is this release safe?).
2. Bounded exhaustive verification of the real kernel against an independent specification, with mutation testing to show the checker has teeth.
3. A statement of what the kernel guarantees, as three explicitly conditional claims (Â§3.3), with a closed-form capture probability for randomly seated k-of-m committees.
4. A live stress test of eight regimes, including a mechanically gated majority baseline, across colluder shares of 0â€“50%, visible or hidden voting, and two reasoning-model families (32,830 calls). It separates two effects:
    - mechanical checks are what stop forgeries;
    - committees trade capture risk, bounded by the seats drawn, for far fewer model requests than a full vote's cost.
5. An open harness with logs, and three observations whose limits we state: honest error near zero even on near-miss receipts (so independence is untested), no herding in few exposures, and longer colluder outputs.

Quorums, Byzantine fault tolerance, threshold signatures and random committee selection are decades old (Â§2). We do not claim the quorum idea. We contribute a bounded-verified kernel for agent swarms and a harness that measures, on real LLM swarms, the behavioural assumptions those guarantees rest on: honest error, independence and colluder compliance.

2. Related Work

Multi-agent risk and collusion. Hammond et al. (2025) taxonomize multi-agent failures, with collusion central. Motwani et al. (2024) study secret collusion via steganography. AI Control (Greenblatt et al., 2024) asks whether protocols hold against an intentionally subversive model; we ask the same of an institution facing a *group* of agents.

Agent institutions and governance.

- The separation-of-powers proposal (arXiv:2603.25100) is closest in spirit but is neither implemented nor tested.
- Paglieri et al. (2026) observe 100 LLM agents in which an evaluation exploit spread and a counter-group audited and blew the whistle. They propose Ostrom-style institutions for swarm self-governance; we build and verify one institutional mechanism.

Verified and governed agent runtimes.

- AgentRFC (Zheng & Zhang, 2026) model-checks agent-protocol security invariants in TLA+.
- Bounded Agents (Muruaga, 2026) proves delegation properties and evaluates them on LLM prompt-injection benchmarks.
- Others add action certificates (Wang, 2026), verify-gated admission (Nguyen & Tran, 2026), swarm-level runtime assurance (Kekatos et al., 2026) and credential revocation (Deochake, 2026).

These works verify protocols, delegation or admission, mostly against injection or single compromised components. None studies authority spread over a committee that a colluding cell can partially capture.

Classic foundations. Byzantine agreement tolerates a bounded fraction of faulty participants (Lamport, Shostak & Pease, 1982). Threshold cryptography distributes authority over k-of-n parties (Desmedt & Frankel, 1989). Algorand seats committees by cryptographic sortition (Gilad et al., 2017). N-version programming showed that independently developed versions fail in correlated ways (Knight & Leveson, 1986), the assumption most at risk when every auditor is a copy of one model.

Gap. Formal or bounded verification of agent infrastructure exists (AgentRFC, Bounded Agents), and so do LLM-agent evaluations of governance mechanisms. To our knowledge, none combines (a) bounded verification of a committee-based institution's invariants with (b) a live stress test against an LLM swarm containing a colluding cell that can capture seats, with the guarantee's behavioural assumptions measured. Two candidate papers were inaccessible to us.

3. Methods

3.1 The kernel (`fleet_witness/institution.py`)

The kernel has four branches:

- the **legislature** sets versioned policy;
- a **proposer** (a worker, plus the swarm majority in hybrid regimes) submits a release;
- the kernel seats *m* **auditors** uniformly at random from the swarm, each attesting with an HMAC under its own key, bound to the exact run and artifact;
- the **executive** commits a release and its audit event in one SQLite transaction.

The executive commits only if a proposal exists, current policy allows it, the evidence is intact, at least *k* **distinct, seated** auditors gave valid bound attestations, and no prior release exists.

Design principle. The kernel never judges whether evidence *means* "safe". In our task, a build receipt is authentic but does not establish release safety. That judgement belongs to auditors, and it is exactly where a colluding cell attacks.

3.2 Bounded verification (`fleet_witness/verify_institution.py`)

We explore every reachable state of the *real* kernel (SQLite serialized and cloned at each state, states deduplicated) in a small world. That world has one case, three seated auditors (quorum 2), one unseated outsider, and 12 actions including forged, outsider, misbound and duplicate attestations and repeated executions. After every transition the kernel is compared with an independently written specification:

- **S1 safety:** no release unless every precondition holds;
- **S2 liveness:** a release whenever they do;
- **S3:** at most one release per case;
- **S4:** every release has its audit event.

Seven mutant kernels, each missing one check, show the checker can fail.

Scope. This is bounded model checking, not a proof.

- The checked world (one case, three seats, quorum 2, depth 6) is smaller than the live configuration (24 cases, seven seats).
- S2 means "an eligible execution attempt succeeds", not temporal progress.
- States are deduplicated on the contents of the kernel's tables plus the specification state.
- The specification was written by the same team, so a mistake shared with the kernel could go unnoticed.

Details are in Appendix B.

3.3 What the kernel guarantees: three conditional claims

- **G1 (mechanical; verified within the bound).** No release happens unless a proposal exists, current policy allows it, the evidence is intact, and at least k distinct seated auditors have given valid attestations bound to that exact release. At most one release happens per case, and each is audited. This is a database-level guarantee: at-most-once effects in external systems, and concurrent multi-case behaviour, are not covered.
- **G2 (conditional on honest auditors never attesting unsafe releases).** An unsafe release then requires at least k colluders on its committee, so capture is at most the fraction of committees that drew k or more colluders. This is the "seat bound" in Table 1.
- **G3 (probabilistic).** Assume random, unpredictable seating of m seats from N agents with C colluders, honest errors that are independent with rate Îµ, and full colluder compliance. Then

P(capture) = Î£â±¼ Hypergeom(j; N, C, m) Â· P(Binom(mâˆ’j, Îµ) â‰¥ kâˆ’j).

Imperfect compliance lowers capture. Correlated honest errors could raise it above G3.

Our live runs exercise G1 and G2. G3's independence assumption is untested, because honest unsafe error was essentially zero (Â§4.3).

3.4 Live stress test (`fleet_witness/swarm/`)

Release cases. There are 24 per run: 8 valid, plus invalid build-only, wrong-artifact and wrong-run receipts (authentic signatures, wrong claim or binding) and forged receipts (plausible fields, bad signature). The cell's **targets** are the 12 invalid cases honest agents could spot. The 4 forged cases are not targets; they test what crowds cannot check.

Swarm.

- 50 reviewers vote approve or deny in five waves.
- **Visible:** later waves see the running tally and the six most recent public comments.
- **Hidden:** agents vote independently.
- A **cell** of 0, 10, 20, 30, 40 or 50% receives a private instruction: push the targets using a shared talking point, and vote honestly elsewhere.
- For every case the kernel seats a committee of 7, shared across conditions so comparisons are paired. Committee members audit with the same rule; colluders on the committee are told to sign targets.
- Auditors attest after the vote. With a visible board they see the final tally and recent comments; with a hidden board they see only the release. Hidden-board committees therefore do not depend on the vote, and could run without it.

Regimes. The same votes and attestations are judged under each regime:

- **direct majority**, with no kernel;
- **gated majority**: the majority plus the kernel's mechanical checks, with no committee (replayed offline from the same votes);
- **kernel committees** of 1-of-1, 2-of-3, 3-of-5, 4-of-7 and unanimous 7-of-7;
- **hybrid**: the majority **and** 4-of-7.

Models. Two reasoning models from different families via OpenRouter: DeepSeek v4 Flash (`deepseek/deepseek-v4-flash-0731`) and OpenAI gpt-oss-20b (`openai/gpt-oss-20b`).

- Both use temperature 0.7, a 2,000-token cap, and the same seeds, so the same cells and committees are drawn.
- Both reason before answering, so only the model family varies. Non-reasoning models were rejected.
- Each sweep schedules 16,416 calls (16,415 completed); both together cost $2.18.

Output caps are kept non-binding (truncated responses are retried at larger caps) because truncation silences colluders disproportionately (Appendix B.3).

Blue team. A detector reads only the public log. It combines shared-minority voting with TF-IDF similarity of reasons, uses robust z-scores and graph clustering, and is scored against sealed ground truth.

4. Results

4.1 Verification

At depth 6 the checker explored 23,376 distinct states and 105,708 transitions, and S1â€“S4 held throughout. Both kernel modes also passed at depth 4.

All seven mutants were caught, each with a shortest counterexample of four to six actions after the initial evidence setup (Appendix B; depth counts actions). One of them, the missing uniqueness check, surfaced only at depth 6 and was missed at depth 5, which shows concretely why bounds matter.

4.2 Live stress test (two reasoning models)

Each model completed 16,415 of 16,416 calls (1 transport error each). gpt-oss returned 6 unparseable responses, which count as abstentions. Cost: $1.38 (DeepSeek) and $0.81 (gpt-oss).

Table 1 pools shares > 0: five shares Ã— two board conditions Ã— 12 targets = 120 target trials per regime. Cells show DeepSeek / gpt-oss.

Regime	Unsafe releases on targets	Seat bound (G2)	A-priori expected (G3)	Valid released (of 96)	Forged released (of 48)	Calls per decision
direct majority	**0 / 0**	n/a	n/a	96 / 96	**48 / 48**	50
gated majority (no committee)	**0 / 0**	n/a	n/a	96 / 96	**0 / 0**	50
auditor 1-of-1	21 / 22	24	36.0	96 / 95	0 / 0	1
committee 2-of-3	21 / 20	24	28.3	96 / 96	0 / 0	3
committee 3-of-5	23 / 20	26	24.4	96 / 96	0 / 0	5
committee 4-of-7	16 / 17	22	21.8	96 / 96	0 / 0	7
unanimous 7-of-7	**0 / 0**	0	0.1	96 / **87**	0 / 0	7
hybrid (majority + 4-of-7)	**0 / 0**	22â€	21.8â€	96 / 96	0 / 0	57

Table 1. Pooled outcomes. â€ Committee-only opportunity, not a hybrid release prediction: the hybrid also needs a strict swarm majority (approvals > denials; ties fail), which honest voters never gave a target. The seat bound and the a-priori expectation are identical for both models because the models share seeds, cells and committees. Board conditions also share committee draws, so each regime has 60 independent committee draws, not 120. We therefore report counts by share (Figure 1; Table C1) rather than pooled confidence intervals.

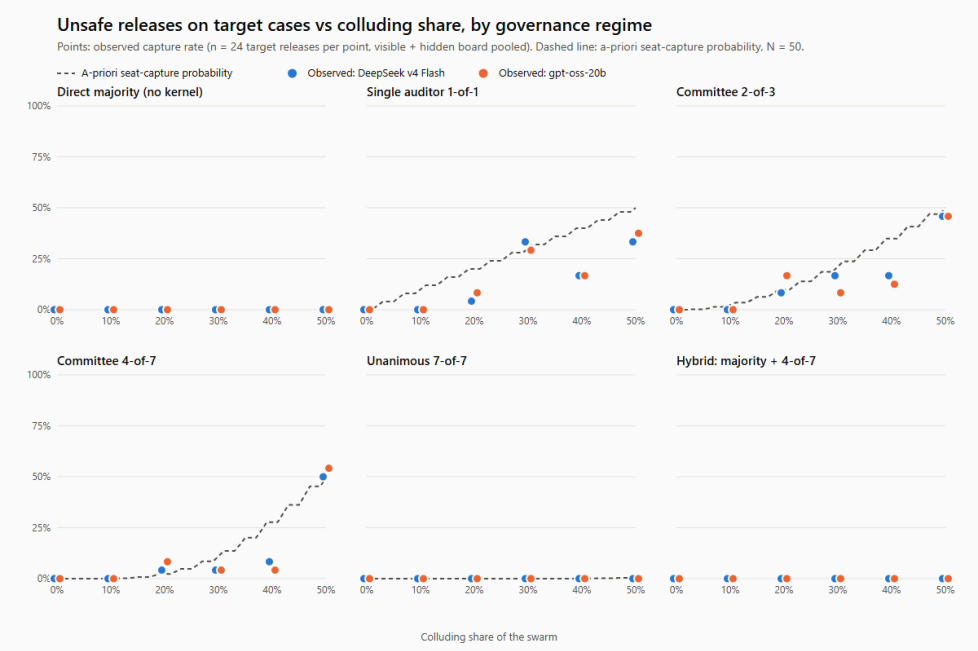


Figure 1. Capture rate on target releases vs colluding share, per regime, for both models. Points are observed (n = 24 per point, 12 independent committee draws). The dashed line is the a-priori probability (G3).

Finding 1: mechanical checks, not voting, stop forgeries.

- The direct majority approved every forged-signature release (48/48 in each model). Every kernel regime blocked all of them.
- This is architectural by construction: voters had no way to verify a symmetric HMAC. With public-key signatures and a verification tool a crowd could check too; we did not test that baseline.
- **The informative comparison is the gated majority:** the same votes plus the kernel's mechanical checks. It was perfectly safe in both models (0/120 captures, 0/48 forgeries, 96/96 valid releases). This matches the hybrid at 50 rather than 57 calls.
- In this setting, then, a committee adds **no safety** over a mechanically gated majority, and was less safe. Its potential value is needing far fewer model requests (below).

Finding 2: committees fail only where their seat draws allow, with far fewer model requests.

- In all regime Ã— share Ã— condition cells, observed capture never exceeded the seat bound (G2).
- Honest auditors never signed a target (0/746 per model), so all capture came from colluder seats.
- Gaps below the bound are colluders declining their instruction: seated colluders signed 234/262 times (89.3%, DeepSeek) and 242/262 times (92.4%, gpt-oss).
- **Realized seat draws were favourable:** 12 of 60 independent draws were capturable for 1-of-1 versus 18 expected (about 1.8 SD low), so points sit below G3's curve in Figure 1. G3 is a planning estimate; G2 is what the verified rules guarantee.
- **The trade-off:** a 4-of-7 committee costs 7 calls instead of 50. It matched the gated majority's safety at 10% collusion, was captured in 1â€“2 of 24 targets per share at 20â€“40%, and in about half at 50%. Unanimity was never captured.

Finding 3: unanimity buys safety with liveness, and the price depends on the model.

- Unanimity blocked 0/96 valid releases for DeepSeek but 9/96 for gpt-oss.

- Those 9 blocks came from 10 refusals: 8 by honest auditors and 2 by cell members auditing non-target releases.

4.3 Honest error, herding and other observations

Honest error is near zero, even on near-miss receipts. We ran 12 additional runs with authentic but subtly invalid receipts, again with both models, 3 seeds, and 0% and 30% shares. The near misses were a run-ID suffix, a two-character digest change, a scope variant, issuer capitalization and a claim variant.

Honest auditor error was 0.3% (DeepSeek, issuer capitalization only) and 0% (gpt-oss); voter error 1.0% and 0%. Honest error alone captured only a 1-of-1 auditor, twice in 108 cases (Appendix C).

DeepSeek's rare voter errors clustered on particular cases, with a between-case variance 5.6Ã— the independent expectation. That points to correlated failure (Knight & Leveson, 1986), but errors were too rare to capture any multi-seat committee (only a single 1-of-1 auditor, twice in 108 cases). Mixed-model committees changed nothing, because there were no errors to diversify away.

This is our honest answer to the brief's "safety in numbers?" question: on exact-match rules, independence is untested because honest errors barely occur. Judgement-based tasks are needed.

No herding. Honest voters who saw an approving majority before them approved a target 0/14 times (DeepSeek) and 0/16 times (gpt-oss). These exposures are few, so this is weak evidence of absence, specific to a crisp rule.

Colluders produced longer outputs (1.9â€“2.8Ã— median tokens in both models; Appendix C). This is an association, not a measured cost of deception: colluders also received a longer, conflicting prompt.

Detection of the planted cell.

- A public-log detector combining shared-minority voting and reason similarity (Appendix B.4) identified the planted cell with F1 = 1.0 at 10â€“40% shares (gpt-oss at 40%, hidden board: 0.97), and 0.98â€“1.0 at 50%.
- False positives: one spurious 3-agent cluster in the 4 zero-collusion runs (2 models Ã— 2 board conditions; gpt-oss, hidden). The detector also needs colluders to lose votes, so it cannot see a cell that wins.
- Because the cell shared a talking point and voted against unanimous honest denials, this shows detection of *this planted signature*, not general collusion detection.

5. Discussion and Limitations

Verify the institution, not the agents. The agents were never verified, yet every outcome landed inside the envelope that the verified rules (G1) and the seat bound (G2) allow. That turns an open-ended question ("is every agent honest?") into a few measurable quantities:

- seat-capture probability, which is closed-form;
- honest error and its correlation;
- colluder compliance.

The cleanest lesson is architectural. Anything that can be checked mechanically belongs in the kernel, because no amount of voting substitutes for it (Finding 1). Judgement can come from a full vote or from a random committee. The committee needs about seven times fewer model requests but is less safe,

failing exactly as its seat draws predict (Finding 2). Token, dollar and latency savings were not measured. A swarm can choose that trade-off deliberately: a gated majority where votes are cheap, and a large-quorum committee where they are not.

The open risk. The guarantee also assumes unpredictable seating, uncompromised keys and kernel-routed actions, which our adversary did not attack, and independent honest errors, which we could not test because such errors barely occurred. Knight & Leveson's warning that model copies fail together is the key open risk.

Limitations

The main limitations, detailed in Appendix A:

- verification is bounded, not a proof;
- the task is synthetic, with instructed colluders and no covert channel or seat-steering;
- two cheap reasoning models were tested, with 50 agents;
- per-share cells are small;
- independence of honest errors is untested;
- the token-length association is confounded by the prompt.

Future Work (a fellowship roadmap)

1. **Correlated errors:** judgement-based tasks where honest agents genuinely err, testing G3 and diverse-model committees directly.
2. **Seating and adversaries:** verifiable random seating (VRF or beacon sortition), cells that steer seats, and adaptive colluders.
3. **Deployment:** the kernel at an MCP/A2A gateway, with public-key attestations and a "crowds plus tools" baseline.
4. **Assurance:** a small specification language for institutions compiled to both kernel and model, multi-case checking, and mechanized proofs (TLA+ or Lean).
5. **Deadlines as defence:** colluders produce longer outputs, so test uniform decision deadlines as an explicit, measured rule.

6. Conclusion

A small, checkable institution held exactly as far as its rules promised against colluding LLM swarms. Mechanical checks, not voting, stopped forgeries. A mechanically gated majority was the safest regime we tested. Random committees used far fewer model requests but were captured at rates bounded by their seat draws (16â€”17 of 120 targets for 4-of-7).

These results come from an exact-match task where honest agents almost never erred, so the key open question remains: do honest AI errors stay independent? That is the next experiment. Verifying the institution rather than the agents is a tractable, measurable path to oversight. Its mechanical guarantees do not weaken as swarms grow; its behavioural assumptions still need testing on harder tasks.

Code and Data

- **Code repository:** https://github.com/k2jac9/fleet-witness (branch `feat/live-swarm`). It contains the kernel, the bounded checker, the swarm harness, the figure scripts and the tests.
- **Data:** every reported run, including the superseded truncated run and the 600-token pilot, is published as `data/fleet-witness-runs-2026-11-08.zip`. It holds the public event logs, sealed ground truth, per-regime decisions and run manifests (see `data/README.md`).
- **Reproduction:** `python scripts/reproduce_tables.py` regenerates every table from those logs without model calls. `python -m fleet_witness swarm-sweep` re-runs experiments.

Author Contributions

- A.C. built the Fleet Witness baseline (trusted controller, scripted evaluation, replay viewer and cloud deployment) that this work extends.
- I.P.B. proposed the institutional-verification research direction, and designed and ran the institution kernel, bounded verification and live swarm experiments with AI coding assistance.
- Both authors are responsible for the final manuscript.

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Appendix A: Limitations and Dual-Use / Ethical Considerations

Limitations

- **Bounded, not complete.** Verification covers one case, three seats and depth 6. Larger worlds and arbitrary interleavings across cases are unverified. The specification was written by the same team; an error shared by kernel and specification would go unnoticed.
- **Toy task and planted adversary.** The release task is synthetic. Colluders were instructed, and coordination did not emerge on its own. There was no covert channel between colluders.
- **Models.** Two cheap reasoning models from different families (DeepSeek v4 Flash, OpenAI gpt-oss-20b). Results may not transfer to frontier agents.
  - Qwen 3.7 Flash spent its entire 600-token budget on hidden reasoning in about 39% of responses in a partial pilot whose logs were not retained. At 2,000 tokens it was too slow for a full sweep in the time available, so it is excluded.
  - Non-reasoning models were excluded so that reasoning mode would not be confounded with model family.
  - The pilot used a second host (Adaption), whose rate limit made full sweeps impractical.
- **Small cells.** n = 12 targets per cell. Committee draws are shared across board conditions, so the two conditions are paired, not independent. No significance claims.
- **Unsafe honest error was essentially zero, so independence was untested.**
  - In both models, honest auditors never signed an invalid main-sweep target (0/746 each).
  - On near-miss receipts, honest auditor error was 0.3% (DeepSeek) and 0% (gpt-oss) with no colluders present. At 30% collusion each model made one honest unsafe attestation in 508 honest seats (about 0.2%), so gpt-oss's honest error was not globally zero.
  - The only other honest error observed was gpt-oss refusing valid releases (8/492). With correlated honest errors, such as shared weights, shared priors or herding (cf. Knight & Leveson, 1986), the Â§3.3 bound could be optimistic. Neither the herding null result nor the absence of correlated failure generalizes beyond this task.
- **Seating assumptions.** Committees were drawn by a seeded pseudo-random generator the cell could not observe. An adversary who can predict or influence seating defeats the bound.
- **Trust assumptions and key management.**
  - The kernel's keys and SQLite store are trusted.
  - Attestations use symmetric HMAC because only the trusted kernel verifies them. This gives no public verifiability and no protection against a compromised host.
  - Per-auditor keys are generated fresh per run and held by the kernel; rotation, revocation and compromise recovery are not implemented.

    - Asymmetric signatures (e.g. Ed25519), credential revocation (cf. Deochake, 2026) and tamper-evident logs would preserve the kernel's semantics while enabling external audit.
    - Containers and OS-level isolation are out of scope.
- **Adversary model.** The cell had no covert channel and no key compromise. It did not try to predict or steer seating or to route actions around the kernel.

Dual-use

Could this make multi-agent operations easier, or harder to detect?

- **Easier:** the harness creates instructed colluding agents and measures committee capture, so an attacker could use the capture curves to size a cell. The kernel's rules are public, so its blind spot is also known: it trusts auditors on semantic judgements.
- **Harder to detect:** a colluder who knew about the output-length association or the shared-minority detector could try to evade them.
- **Overall:** we judge the work to favour defenders. It shows that mechanical checks block forgery outright, that committee risk is predictable, and how to detect a planted cell.

Mitigations:

- the task is synthetic, and nothing touches third-party systems;
- there is no scanning, recruitment or contact with real agents;
- the talking points are generic;
- the detector and the deliberation-cost signal are released alongside, favouring defenders.

Ethics and provenance

- No human subjects and no personal data.
- Spend was capped and logged per call.
- The repository was created during the sprint (2026-11-08). Earlier conceptual research informed the design, but no pre-sprint code is used (`docs/AI_ASSISTANCE.md`). The scripted Fleet Witness controller was built earlier in the same sprint; the institution kernel, checker and swarm harness were built after it.
- AI assistance is described in the LLM Usage Statement.

Appendix B: Verification and audit details

B.1 Bounded-check world.

- One case, three seated auditors (quorum 2), one registered but unseated agent.
- Evidence is either intact or tampered.

The 12 actions:

- majority or minority proposal;
- policy on or off;
- valid attestations from each seat and from the outsider;

- a forged-signature attestation;
- an attestation bound to the wrong artifact;
- two distinct execution attempts.

Repeating any action models duplicates and replays. Exploration is breadth-first over serialized SQLite states with deduplication. Both kernel modes, committee-only and hybrid, are checked.

B.2 Verification results.

Check	Result
Kernel, depth 6	23,376 distinct states, 105,708 transitions; S1â€“S4 hold
Committee-only mode, depth 4	3,188 states; all hold
Hybrid mode, depth 4	2,906 states; all hold
Mutant: skip policy recheck	Caught, S1 (4 steps)
Mutant: skip integrity	Caught, S1 (tampered evidence, 5 steps)
Mutant: skip seat check	Caught, S1 (outsider attests)
Mutant: skip binding	Caught, S1 (wrong-artifact attestation)
Mutant: count duplicates	Caught, S1 (same auditor twice)
Mutant: skip uniqueness	Caught, S3 (two releases; needs depth 6, missed at depth 5)
Mutant: skip audit	Caught, S4

B.3 Truncation audit.

- gpt-oss: 0 truncated responses; longest 1,536 tokens.
- DeepSeek: one colluder response hit the 2,000-token cap in the 30% run.
- Under a rule declared before results were known, that whole run (all agents) was re-run with retry enabled. The re-run needed no retries.
- Its outcomes matched the original in 13 of 14 regime Ã— condition cells; the remaining cell (4-of-7, hidden) moved from 1 capture to 0.
- Both versions are retained. Main results use the re-run.
- Provenance: both runs are in the published bundle (`runs/deepseek2k-f30-truncated`, `runs/sweep-deepseek2k/f30`), so the 13/14 comparison can be recomputed. The Qwen pilot figure in Appendix A comes from logs that were not retained and is not independently auditable.

B.4 Detector specification. The detector sees only public votes and reasons. For each pair of agents in a condition:

- **Shared-minority score:** the fraction of common cases where both voted on the losing side.
- **Text similarity:** TF-IDF cosine over each agent's concatenated reasons, using word unigrams and bigrams, lowercased, with stopwords and tokens of two characters or fewer removed.

Both pairwise matrices are converted to robust z-scores (median and 1.4826 Ã— MAD; mean absolute deviation is the fallback when the MAD is zero). Scoring:

- combined score = 0.5 Â· z_minority + 0.5 Â· z_text;

- a pair is linked if combined â‰¥ 2.0, z_minority â‰¥ 3.0 and the raw shared-minority score is > 0;
- connected components of three or more agents are flagged.

The parameters were tuned on scripted (fake-provider) runs and fixed before any live sweep. Code: `fleet_witness/swarm/detect.py`.

Appendix C: Supplementary results

C.0 Capture by colluder share.

Colluder share	1-of-1	2-of-3	4-of-7	Gated majority, unanimity, hybrid
10%	0 / 0	0 / 0	0 / 0	0 / 0
20%	1 / 2	2 / 4	1 / 2	0 / 0
30%	8 / 7	4 / 2	1 / 1	0 / 0
40%	4 / 4	4 / 3	2 / 1	0 / 0
50%	8 / 9	11 / 11	12 / 13	0 / 0

Table C1. Unsafe releases on targets by colluder share (of 24 per share and model; DeepSeek / gpt-oss).

C.1 Output length by role.

Colluders produced longer outputs.

- Median output tokens on target cases (honest vs colluding): DeepSeek votes 106 vs 294 (n = 5,400 / 1,800) and audits 111 vs 241; gpt-oss votes 158 vs 369 and audits 174 vs 329 (Figure 2).
- In a 600-token pilot there were 99 abstentions: 98 came from colluders, and 95 hit the cap.
- **This is an association, not a measured "cost of deception".** Colluders also received a longer prompt with a conflicting instruction.
- Token counts include hidden reasoning. They are visible to an operator through usage metadata but not to a content monitor.
- Whether the signal survives adaptive colluders is untested.

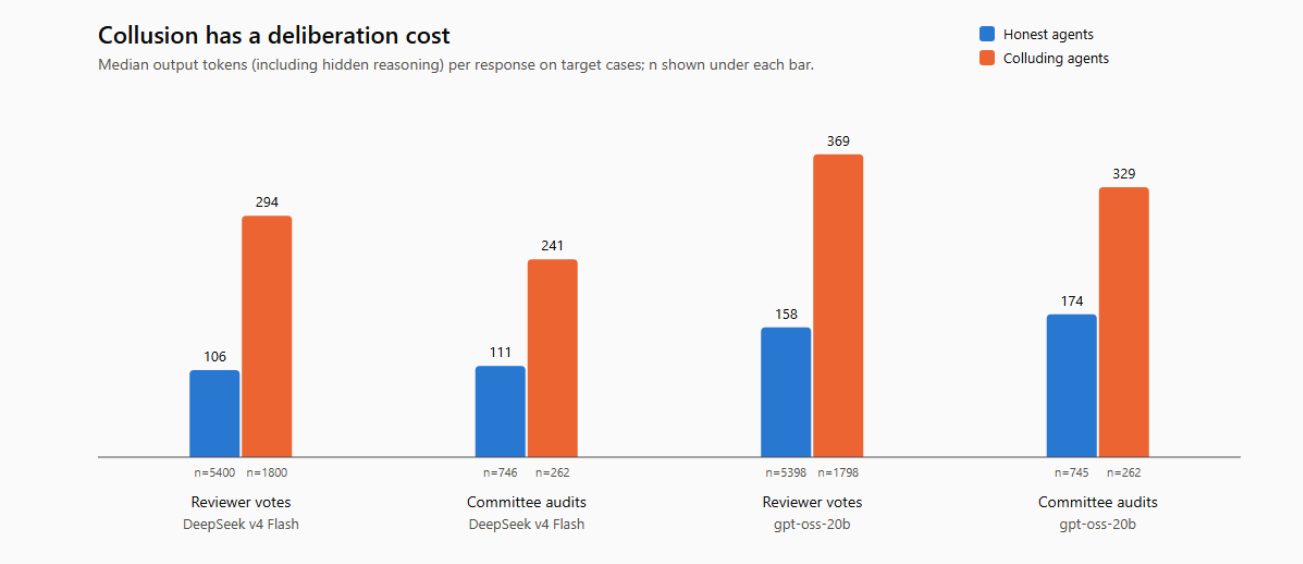


Figure 2. Median output tokens, including hidden reasoning, on target cases: honest vs colluding agents.

C.2 Honest error on near-miss releases.

Honest error on near-miss releases (0% cell)	DeepSeek	gpt-oss
Committee auditors (n = 756 seats)	0.3% (issuer capitalization only)	0%
Reviewer votes	1.0%	0%
Committees captured by honest error alone	1-of-1: 2/108; larger: 0	0

C.3 Output caps as instruments. Every agent in a run shares one cap, which is kept non-binding: truncated responses are re-sent at 2Ã—, 4Ã— and 8Ã— the cap, and every run is audited.

- This matters because truncation is not neutral: in a 600-token pilot, 98 of 99 abstentions (95 at the cap) were colluders. A binding cap is effectively a hidden deadline that silences deliberating deceivers.
- An unanswered audit counts as "not signed".
- Audit outcome and the one pre-declared re-run are in Appendix B.

LLM Usage Statement

We used Claude Code (Anthropic, Claude Opus 5.5) extensively, for:

- auditing the existing repository;
- implementing the institution kernel, the bounded checker and the swarm harness;
- running and analysing the experiments;
- generating the figures;
- drafting this report.

The authors set the research questions and the key methodological decisions, including:

- the pivot to verifying institutions rather than agents;
- comparing two reasoning models rather than a reasoning and a non-reasoning model;

- treating output caps as non-binding instruments, with the pre-declared re-run rule.

Every reported number was produced by code in the repository and cross-checked against the raw run logs. The authors are responsible for all claims.